



Project Plan: Multi-Boardroom Booking System for FlexiSpace

Project Overview

The objective is to implement a centralized boardroom booking system that enables customers and staff to reserve meeting rooms across multiple FlexiSpace locations while preventing double bookings through real-time synchronization with Microsoft Outlook calendars used by Centre Managers.

The solution will support the following locations:

Location	Centre Manager	Outlook Calendar
Centurion	Centre Manager	centurion@flexispace.net.za
Houghton	Rebecca	rebecca@flexispace.net.za
Eagle Canyon	Antoinette & Lesego	antoinette@flexispace.net.za , lesego@flexispace.net.za (assuming the intended address is lesego@flexispace.net.za)

Project Objectives

The system will:

- Provide one booking platform for all FlexiSpace locations.
- Display live boardroom availability.
- Synchronize with Outlook calendars.
- Prevent double bookings.
- Allow Centre Managers to manage bookings.
- Generate booking confirmations.
- Record booking history.
- Support future office locations.

Scope

Included

Office Locations

- Centurion
- Houghton
- Eagle Canyon

Boardrooms

Each location may have multiple boardrooms.

Example:

Centurion

- Boardroom A
- Boardroom B
- Training Room

Houghton

- Boardroom 1
- Executive Boardroom

Eagle Canyon

- Confuzzled
- Thingamajik
- Whachamacallit
- Fiddlestix
- Meeting Room
- Training Room

(Final inventory to be confirmed.)

Users

Administrators

Can:

- Add/remove boardrooms
 - Configure locations
 - Manage users
 - View all bookings
 - Generate reports
-

Centre Managers

Can:

- View bookings
 - Approve bookings (optional)
 - Block rooms
 - Cancel bookings
 - Edit bookings
-

Staff

Can:

- Book rooms
 - Cancel own bookings
 - View availability
-

Clients (Optional Phase 2)

Can:

- Book meeting rooms online
 - Receive confirmations
 - Pay online (future enhancement)
-

Functional Requirements

1. Dashboard

Displays:

- Today's bookings
 - Current room availability
 - Upcoming meetings
 - Occupancy statistics
 - Location overview
-

2. Booking Calendar

Features:

- Day view
- Week view
- Month view
- Timeline view

Filters:

- Location
 - Boardroom
 - Capacity
 - Equipment
-

3. Room Availability

Real-time availability showing:

Available

Occupied

Reserved

Cleaning

Maintenance

Blocked

4. Outlook Calendar Integration

Microsoft 365 Integration

Each booking must synchronize with Outlook.

Eagle Canyon

Monitor:

- antoinette@flexispace.net.za
- lesego@flexispace.net.za

Houghton

Monitor:

- rebecca@flexispace.net.za

Centurion

Monitor:

- centurion@flexispace.net.za

Synchronization Rules

When Outlook changes:

- Booking updated
- Booking cancelled
- Meeting moved

System updates automatically.

When system changes:

- Outlook calendar updated immediately.
-

5. Conflict Detection

Before saving:

System checks:

- Room availability

- Outlook calendar
- Existing bookings
- Blocked periods

If conflict exists:

Booking rejected.

Alternative times suggested.

6. Booking Details

Each booking records:

- Booking ID
- Date
- Start time
- End time
- Location
- Boardroom
- Booker
- Company
- Number of attendees
- Catering required
- Equipment required
- Notes
- Outlook Meeting ID

7. Equipment

Track:

- TV
- Projector
- Whiteboard

- Video conferencing
 - Teams Room
 - Flip chart
 - HDMI cables
-

8. Catering

Optional:

Tea

Coffee

Lunch

Snacks

Water

9. Notifications

Automatic emails for:

Booking created

Booking cancelled

Booking modified

Reminder (24 hrs)

Reminder (1 hr)

10. Reports

Reports include:

Daily bookings

Monthly usage

Occupancy %

Most used rooms

Cancelled bookings

Revenue (future)

Outlook Integration Design

Microsoft Graph API

Recommended integration:

Microsoft Graph API

Functions:

- Read calendars
 - Create meetings
 - Update meetings
 - Delete meetings
 - Check availability
-

Calendar Synchronization Flow

User books room

↓

System checks Outlook Calendar

↓

Room available?

↓

YES

↓

Create booking

↓

Create Outlook Event

↓

Send confirmation

↓

Calendar updated

Synchronization Frequency

Preferred:

Real-time using Microsoft Graph Webhooks.

Fallback:

Every 2 minutes.

Database Design

Locations

- LocationID
 - Name
 - Address
-

Boardrooms

- RoomID
 - Name
 - Capacity
 - LocationID
 - Equipment
-

Bookings

- BookingID
 - RoomID
 - Start
 - End
 - User
 - Status
 - OutlookEventID
-

Users

- UserID
 - Name
 - Email
 - Role
 - Location
-

Equipment

- EquipmentID
 - Description
-

Security

Authentication:

Microsoft 365 login

Permissions:

Administrator

Centre Manager

Staff

Clients

Audit logs for:

- Booking changes
- Deletions
- Login history

User Interface

Home

Locations

[Centurion]

[Houghton]

[Eagle Canyon]

Calendar

Monday

09:00

Boardroom A



Boardroom B

Available

Training Room



Booking Screen

Select:

Location



Boardroom



Date



Time



Attendees



Equipment



Submit

Technology Stack

Front End

- React
- Microsoft Fluent UI

Backend

- .NET 8 Web API
- Microsoft Graph API

Database

- Microsoft SQL Server

Authentication

- Microsoft Entra ID (Azure Active Directory)

Hosting

- Microsoft Azure
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Implementation Phases

Phase	Duration	Deliverables
Discovery & Requirements	1 week	Requirements, room inventory, workflow mapping
Solution Design	1 week	Architecture, database, UI mockups
Microsoft 365 & Graph Integration	2 weeks	Calendar synchronization, authentication, webhook setup
Core Booking Engine	2 weeks	Booking logic, conflict detection, availability engine
User Interface Development	2 weeks	Web application, calendar views, booking forms
Notifications & Reporting	1 week	Email reminders, reporting dashboards
System Testing (UAT)	1 week	Functional testing, Outlook sync validation, user acceptance

Phase	Duration	Deliverables
Training & Go-Live	1 week	User training, production deployment, support handover

Estimated Project Duration: 10–11 weeks

Success Criteria

The project will be considered successful when:

- No double bookings occur across any location.
- Outlook and the booking system remain synchronized in near real time.
- Users can search and book available boardrooms in under one minute.
- Centre Managers can manage all bookings from a single interface.
- Automated notifications reduce manual coordination.
- The platform can be extended to additional FlexiSpace locations with minimal configuration.

Future Enhancements

- Visitor management integration.
- QR code check-in at boardrooms.
- Digital room signage outside meeting rooms.
- Microsoft Teams and Zoom meeting creation.
- Catering workflow with supplier notifications.
- Power BI dashboards for utilization and occupancy analytics.
- Mobile application (iOS and Android).
- Customer self-service portal with online payments.
- Integration with CRM and billing systems for chargeable bookings.